

DHRUV ARORA

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PROFESSIONAL SUMMARY

Mathematics graduate (CGPA: 8.34, Minor: Computer Science) from Hansraj College, University of Delhi, with hands-on experience building end-to-end ML pipelines and data analytics dashboards. Proficient in Python, Scikit-learn, SQL, Power BI, and Excel — across the full workflow of data cleaning, EDA, feature engineering, model development, evaluation, and visualization. Built classification and regression systems achieving 88.2% accuracy and $R^2=0.77$ respectively. Applied exposure to deep learning architectures (FNN, CNN, RNN).

EDUCATION

B.Sc. (Hons) Mathematics, Minor in Computer Science Hansraj College, University of Delhi CGPA: 8.34	2022 – 2026
Senior Secondary – Class XII (CBSE) Bal Bharati Public School Percentage: 92.2%	2021 – 2022

TECHNICAL SKILLS

Programming & Databases: Python, SQL (SELECT, JOINS, CRUD, Stored Procedures)

ML Frameworks & Libraries: Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning: Linear/Logistic Regression, Lasso/Ridge, Decision Trees, Random Forest, Gradient Boosting (XGBoost), KNN, Naive Bayes, SVM, K-Means Clustering

Deep Learning: Neural Networks (FNN, CNN, RNN), Reinforcement Learning

ML Engineering: Pipeline Design (Scikit-learn), Feature Engineering, Hyperparameter Tuning (GridSearchCV), Cross-Validation, Model Evaluation (Accuracy, F1, ROC-AUC, R^2)

Business Intelligence: Power BI (DAX, Data Modelling, Star Schema, KPI Cards, Slicers), Excel (Pivot Tables, Charts, Data Cleaning)

Data Analysis: EDA, Statistical Modelling, Correlation & Regression Analysis, Hypothesis Testing

Mathematical Foundations: Probability, Statistics, Linear Algebra, Distributions (Binomial/Poisson/Normal)

Tools: Jupyter Notebook, Git, GitHub, VS Code, MS Excel, Power BI

PROJECTS

CreditWise Loan Approval System

github.com/dhruvarora2004/CreditWise-Loan-Approval-System

Tools: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn)

- Architected Scikit-learn Pipelines with StandardScaler for Logistic Regression, Naive Bayes, and KNN; tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Engineered features including squared Credit Score and DTI Ratio terms; handled missing values via domain-derived contextual imputation using np.select
- Naive Bayes selected as best model — Accuracy: 88.2%, F1: 0.80, ROC-AUC: 0.84 — recommended for production based on business-context trade-off analysis
- Designed modular src/ architecture; production-ready structure with clean reproducibility via single entry point

Student Exam Performance Analyzer

github.com/dhruvarora2004/student-performance-analysis

Tools: Python (Pandas, Scikit-learn, Matplotlib, Seaborn)

- Compared 7 regression models (Linear Regression, Lasso CV, Ridge CV, KNN, Decision Tree, Random Forest, Gradient Boosting) with GridSearchCV and 5-fold CV on 6,607 student records
- Linear models achieved $R^2=0.77$, outperforming all ensemble methods (Gradient Boosting: 0.74, Random Forest: 0.69); confirmed predominantly linear feature space
- Caught and fixed a label inversion bug in manual ordinal encoding that was distorting correlations for parental involvement, resource access, and teacher quality
- Lasso CV identified Access to Resources and Parental Involvement as top predictors, outweighing study hours

Sales Performance Analytics Dashboard

Tools: Python (Pandas, Matplotlib, Seaborn), Excel (Pivot Tables), Power BI

- Analyzed 200K retail transactions (2023–2024) across 4 regions, 4 categories, and 19 sub-categories; engineered Profit Margin % feature and parsed datetime for time-series analysis
- Identified Electronics as top revenue category (~₹57M), November as seasonal peak, consistent YoY growth; built 3 Excel Pivot Tables with supporting charts
- Built interactive Power BI dashboard (star schema, DAX measures) with KPI cards (Revenue: ₹142M, Profit: ₹32M), dynamic slicers, 24-month trend line, and state-level map visual

Statistical Analysis – Microsoft Excel

Hansraj College

Guide: Prof. Kapil Kumar, Department of Mathematics

- Applied binomial, Poisson, and normal probability distributions along with regression analysis on real-world datasets
- Used Excel Pivot Tables, charts, and visualization techniques to derive and communicate statistical insights

WORK EXPERIENCE

Mathematics Tutor – Home Tuition

2022 – 2025

- Delivered personalized mathematics coaching to 10–25 students, tracking individual performance to identify learning gaps and adapt teaching strategies
- Developed structured problem-solving frameworks that improved students' conceptual clarity and exam scores across academic levels

CERTIFICATIONS & COURSES

- McKinsey Forward Program — Modules: Adaptability & Resilience, Problem Solving, Communicating for Impact, Relationships & Well-Being, Digital Toolkit
- AI/ML & Data Analytics – Apna College
- Data Analytics and Visualization with MS Power BI – IIM Bangalore (IIMB) via Swayam
- Data Analytics with Python – NPTEL via Swayam

LANGUAGES

English (Fluent) | Hindi (Native)